Michael Solomentsev
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Education

PhD in Electrical and Computer Engineering — August 2019 – December 2023
The University of Texas at Austin
Advisor: Professor Alex Hanson
Dissertation: Design & Optimization of Magnetic Components for High-Frequency Power Electronics

B.S. in Electrical and Computer Engineering — August 2015 – May 2019
Cornell University – Magna Cum Laude

Experience:

CEO/Founder — July 2024 - Present
Palanquin Power, Inc. (Golden, CO)

- Led effort to commercialize ultra-high efficiency differential power processing architectures for data center applications, based on my PhD research.
- Engineering, fundraising, business development, and everything else.
- Demonstrated highest power differential power processing system ever built.
- Designed and prototyped DC/DC power converters (DAB topology, from simulation to PCB design to hardware bringup to system integration).
- Managed and built team of stakeholders from Palanquin, UT Austin, & NREL.
- Attracted ~$1 million in non-dilutive funding to the company [NREL West Gate Lab-Embedded Entrepreneurship Award (~$400k); NSF SBIR Phase I ($305k); Colorado Office of Economic Development ($250k)]
- Won several pitch competitions (NREL Industry Growth Forum 2025 – Best Early Stage Venture; Open Compute Project Future Technologies Summit 2025).

Postdoctoral Fellow — January 2024 – July 2024
Power Electronics & Magnetics Group – UT Austin

- Supervised graduate researchers working on power magnetics projects.
- Participated in NSF I-Corps national cohort, conducting over 100 customer discovery interviews in the data center market.
- Worked to commercialize innovative research.

Fellow — June 2018 – July 2024
PRIME Coalition (Cambridge, MA)

- Modeled climate impact of early-stage clean tech startups
- Consulted on industry-wide standardization of impact metrics & evaluation of GHG emissions reduction
- Scouted and evaluated companies for inclusion in PRIME Impact Fund
- Met with entrepreneurs to evaluate suitability for charitable capital

Research Assistant — August 2019 – December 2023
Power Electronics & Magnetics Group – UT Austin

- Developed methods for design, analysis, and optimization of high frequency inductors and transformers.

- Projects developing differential power processing architectures for photovoltaics, power line energy harvesting, wireless power transfer, novel magnetic structures.
- Extensive work with PCB design, FEA simulation, circuit simulation tools.
- Supervised several undergraduate and graduate researchers.

Undergraduate Research Assistant August 2018 – May 2019
Collective Embodied Intelligence Lab – Cornell University

- Implemented computer vision solutions for early season vineyard yield estimation.
- Built sensor testing rig for new angle sensitive pixel device.

Algorithm & Applications Intern June – August 2017
Kionix Inc. (Ithaca, NY)

- Identified bugs in operation of ASICs for the company's newest sensor products.
- Developed novel applications, including magnetometer-based door position detection system.
- Performed validation tests. Assisted completion of linearity testing protocol for accelerometers.

Publications - Journal:

[J1] A. Brown, **M. Solomentsev** and A. J. Hanson, "Parallel Resonant Loss Characterization of High Frequency Magnetic Components," in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, doi: 10.1109/JESTPE.2024.3463537.

[J2] **M. Solomentsev** and A. J. Hanson, "At What Frequencies Should Air-Core Magnetics Be Used?," in IEEE Transactions on Power Electronics, vol. 38, no. 3, pp. 3546-3558, March 2023, doi: 10.1109/TPEL.2022.3222993.

[J3] **M. Solomentsev** and A. J. Hanson, "Modeling Current Distribution Within Conductors and Between Parallel Conductors in High-Frequency Magnetics," in IEEE Open Journal of Power Electronics, vol. 3, pp. 635-650, 2022, doi: 10.1109/OJPEL.2022.3212903.

Publications - Conference:

[C1] P. Saraf, **M. Solomentsev** and A. Hanson, "A ZCS-ZVS Strategy For Low Impedance Dual Active Bridges in MHz range," 2025 IEEE Applied Power Electronics Conference and Exposition (APEC), Atlanta, GA, USA, 2025, pp. 77-83, doi: 10.1109/APEC48143.2025.10977490.

[C2] A. Brown, **M. Solomentsev**, C. Fu, O. Okeke and A. J. Hanson, "Double Sided Conduction in N:1 Transformers," 2024 IEEE Applied Power Electronics Conference and Exposition (APEC), Long Beach, CA, USA, 2024, pp. 884-889, doi: 10.1109/APEC48139.2024.10509440.

[C3] **M. Solomentsev** and A. J. Hanson, "Highly-Scalable Differential Power Processing Architecture for On-Vehicle Photovoltaics," 2023 IEEE 24th Workshop on Control and Modeling for Power Electronics (COMPEL), Ann Arbor, MI, USA, 2023, pp. 1-7, doi: 10.1109/COMPEL52896.2023.10220974.

[C4] A. Nguyen, A. Phanse, **M. Solomentsev**, A.J. Hanson, "A low-leakage, low-loss magnetic transformer structure for high-frequency applications", EPE 2022 ECCE Europe, Hanover (2022).

[C5] **M. Solomentsev**, A. Bendapudi and A. Hanson, "QuickSHiFT: Rapid High-Frequency Transformer Simulation and Optimization", IEEE Workshop on Control and Modeling for Power Electronics (COMPEL), June 2022.

[C6] **M. Solomentsev** and A. Hanson. “At what frequencies should air core magnetics be used?” IEEE Applied Power Electronics Conference, March 2022. *__Best Presentation Award.__*

[C7] **M. Solomentsev**, O. Okeke, and A. Hanson. “A Resonant Approach to Transformer Characterization” IEEE Applied Power Electronics Conference, March 2022.

[C8] O. Okeke, **M. Solomentsev**, and A. Hanson. “Double-Sided Conduction: A Loss-Reduction Technique for High Frequency Transformers” IEEE Applied Power Electronics Conference, March 2022.

[C9] **M. Solomentsev** and A. Hanson. “Modeling Current Distribution Within Conductors and Between Parallel Conductors in High Frequency Transformers.” IEEE Applied Power Electronics Conference, June 2021. *__Best Presentation Award.__*

Patents:

[P1] US Patent Application No. 63/644,472, filed May 08, 2024, titled “Scalable, Buffered Differential Power Processing Architecture”.

Invited Talks:

- Panelist - ‘Rack Infrastructure’; Texas Initiative for Datacenter Energy and Large Loads (TEX-DEL) Workshop; November 11-12, 2025; Austin, TX
- ‘Data Center Power Infrastructure & Issues’; Texas A&M University Public Service & Administration Capstone Course; September 26, 2025; Virtual.
- ‘Differential Power Processing for Ultra-Efficient Rack Level Power Conversion’; Open Compute Project EMEA Summit; April 29, 2025; Dublin, IE.

Teaching & Service:

Curriculum Development Assistant — Summer 2021
UT Austin - EE302: Introduction to Electrical Engineering

- Developed lab curriculum for revamped version of introductory course curriculum.
- Tested and refined curriculum based on student feedback and course objectives.

Teaching Assistant — Fall 2018
Cornell University - ECE3400: Intelligent Physical Systems

- Ran lab sessions, guest lectured on additive manufacturing, and milled PCBs.

Lab Manager — May 2018 – May 2019
Cornell University Maker Club

- Handled budgeting, training, and operations for the university’s only student-run makerspace. Organized hardware hackathon for 100+ students.

Honors & Awards:

- NREL West Gate Lab-Embedded Entrepreneurship Program (2024-26)
- Thrust 2000 Graduate Fellowship (2021-24)
- NSF Innovation at the Nexus of Food/Energy/Water Systems (INFEWS) Scholar (2019-21)
- Meinig Family National Scholar (2015-19)
- Kessler Fellowship (2018)